

- Ensemble Method

Week Learner 1

Week Learner 2

Strong Learner

Week Learner 3

Week Learner 4

...

→ Ensemble

① Bagging : รวมกันหลายๆ

② Boosting : ฝึกแบบวน

③ Stacking : ใช้ Meta Learner (ตัวคุม) มาควบคุม

→ Bagging : Leo Breiman (2001)



→ Boosting

→ Stacking

Bootstrap Aggregating

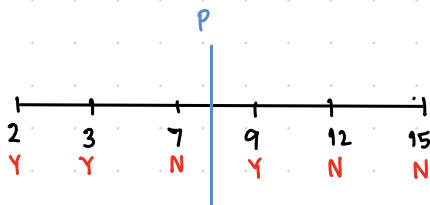
- Random forest [Breiman (2001)]

→ 1400 trainer 1070 train data 330 test

→ 4000 data 900 900 trainer vote

การ Split Data (Continuous) กับ Decision Tree

1. ตัวอย่าง



$$\text{Info}(D) = - \left(\frac{3}{6} \log_2 \left(\frac{3}{6} \right) + \frac{3}{6} \log_2 \left(\frac{3}{6} \right) \right)$$
$$= 1$$

$$\text{Info}_P(D) = \frac{3}{6} I(2,1) + \frac{3}{6} I(1,2)$$
$$= \frac{3}{6} (0.9183) + \frac{3}{6} (0.9183)$$
$$= 0.9883$$

$$\text{Gain}_P = 1 - 0.9883$$
$$= 0.0117$$